Unit 7.01: Logistic Regression & Classification

Q1. In evaluating a binary classifier, the confusion matrix shows TP = 45 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[x] A: 81.0%
[ ] B: 81.8%
[ ] C: 52.4%
[ ] D: 13

Solution: Accuracy is the ratio of correct predictions to total samples: text{Accuracy} = frac{TP + TN}{TP + TN + FP + FN} = frac{45 + 40}{105} = frac{85}{105} = 81.0% .

Q2. In evaluating a binary classifier, the confusion matrix shows TP = 50 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[x] A: 81.8%
[ ] B: 83.3%
[ ] C: 54.5%
[ ] D: 13

Solution: Accuracy is the ratio of correct predictions to total samples: text{Accuracy} = frac{TP + TN}{TP + TN + FP + FN} = frac{50 + 40}{110} = frac{90}{110} = 81.8% .

Q3. In evaluating a binary classifier, the confusion matrix shows TP = 55 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[x] A: 82.6%
[ ] B: 84.6%
[ ] C: 56.5%
[ ] D: 13

Solution: Accuracy is the ratio of correct predictions to total samples: text{Accuracy} = frac{TP + TN}{TP + TN + FP + FN} = frac{55 + 40}{115} = frac{95}{115} = 82.6% .

Q4. In evaluating a binary classifier, the confusion matrix shows TP = 60 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[x] A: 83.3%
[ ] B: 85.7%
[ ] C: 58.3%
[ ] D: 13

Solution: Accuracy is the ratio of correct predictions to total samples: text{Accuracy} = frac{TP + TN}{TP + TN + FP + FN} = frac{60 + 40}{120} = frac{100}{120} = 83.3% .

Q5. In evaluating a binary classifier, the confusion matrix shows TP = 65 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[x] A: 84.0%
[ ] B: 86.7%
[ ] C: 60.0%
[ ] D: 13

Solution: Accuracy is the ratio of correct predictions to total samples: text{Accuracy} = frac{TP + TN}{TP + TN + FP + FN} = frac{65 + 40}{125} = frac{105}{125} = 84.0% .

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Q6. In evaluating a binary classifier, the confusion matrix shows TP = 70 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[x] A: 84.6%
[ ] B: 87.5%
[ ] C: 61.5%
[ ] D: 13

Solution: Accuracy is the ratio of correct predictions to total samples: $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 40}{130} = \frac{110}{130} = 84.6\%$.

Q7. A spam classification model identifies TP = 30 genuine spam emails and FP = 10 legitimate emails falsely flagged as spam. What is the precision of this model?

[x] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%

Solution: Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30 + 10} = \frac{30}{40} = 75.0\%$.

Q8. A spam classification model identifies TP = 60 genuine spam emails and FP = 20 legitimate emails falsely flagged as spam. What is the precision of this model?

[x] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%

Solution: Precision is the fraction of predicted positive instances that are truly positive: $\text{Precision} = \frac{TP}{TP + FP} = \frac{60}{60 + 20} = \frac{60}{80} = 75.0\%$.